Physics 5313: Electromagnetic Theory 2 (Fall 2026)

Lecture 07: EM waves at the surface of a conductor

Instructor: Zihan Wang

Index of refraction in a conductor ($\omega \ll \gamma_0$)

$$\varepsilon(\omega) = \varepsilon_r + i\frac{b}{\omega}$$

$$n = \sqrt{\frac{\varepsilon\mu}{\varepsilon_0\mu_0}} = c\sqrt{\mu\varepsilon}$$ assume μ is a real number

$$n^2 = c^2\mu\varepsilon \qquad n = n_r + in_i$$

$$\Rightarrow \quad (n_r + in_i)^2 = c^2\mu\left(\varepsilon_r + i\frac{b}{\omega}\right)$$

$$\begin{cases} n_r^2 - n_i^2 = c^2\mu\varepsilon_r \\ 2n_i n_r = \dfrac{c^2\mu b}{\omega} \end{cases}$$

Index of refraction in a conductor ($\omega \ll \gamma_0$)

$$\begin{cases} n_r = c\sqrt{\frac{\mu\varepsilon_r}{2}}\left[\sqrt{1+\left(\frac{\sigma}{\varepsilon_r\omega}\right)^2}+1\right]^{1/2} \\ n_i = c\sqrt{\frac{\mu\varepsilon_r}{2}}\left[\sqrt{1+\left(\frac{\sigma}{\varepsilon_r\omega}\right)^2}-1\right]^{1/2} \end{cases}$$

$\sigma \to 0$, insulator. $n_i = 0$, $n_r = c\sqrt{\mu\varepsilon_r}$

$\sigma \gg \varepsilon_r\omega$, $n_r = n_i = c\sqrt{\frac{\mu\sigma}{2\omega}} \gg c\sqrt{\frac{\varepsilon_r\mu}{2}} \approx \frac{\sqrt{2}}{2}$

Good conductor

Propagation in a conductor ($\omega \ll \gamma_0$)

$$\vec{E} = \vec{E}_0 e^{i(\vec{k}\cdot\vec{x} - \omega t)}$$

$$\vec{H} = \vec{H}_0 e^{i(\vec{k}\cdot\vec{x} - \omega t)}$$

$$\vec{k} = \frac{n\omega}{c}\hat{k} = (n_r + i n_i)\frac{\omega}{c}\hat{k}$$

$$\vec{E} = \vec{E}_0 e^{i\frac{\omega}{c}\hat{k}\cdot\vec{x}(n_r + i n_i) - i\omega t}$$

$$= \vec{E}_0 e^{-\frac{\omega}{c} n_i \hat{k}\cdot\vec{x}} e^{i\left(\frac{\omega}{c} n_r \hat{k}\cdot\vec{x} - \omega t\right)}$$

Decay: $\frac{\omega}{c} n_i = \frac{\omega}{c} c\sqrt{\frac{\mu\sigma}{2\omega}} = \sqrt{\frac{\omega\mu\sigma}{2}}$

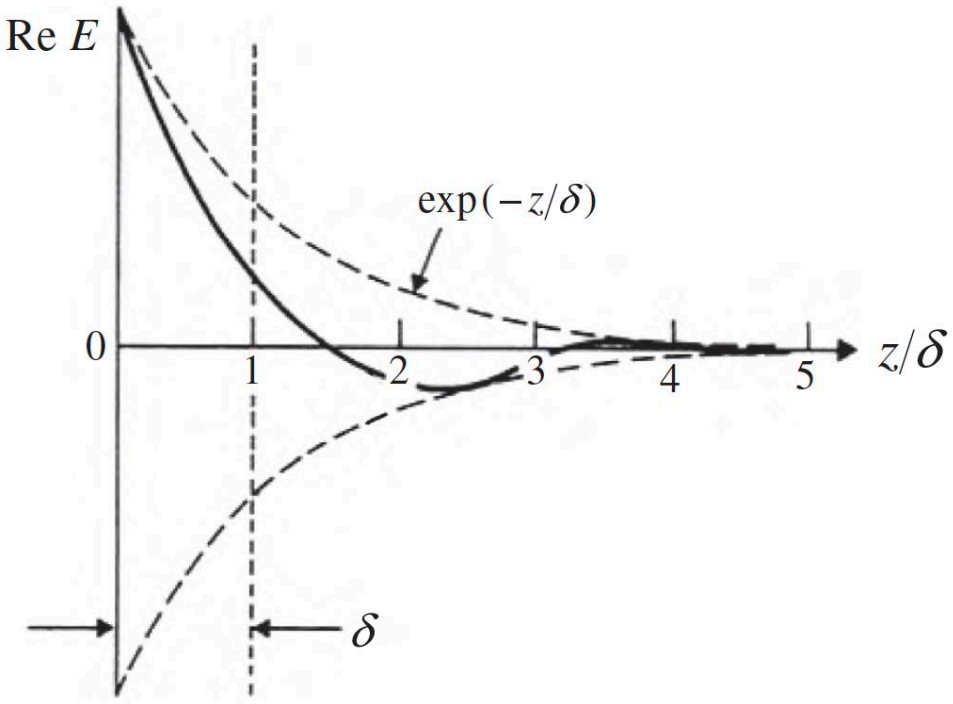
$\delta = \sqrt{\frac{2}{\mu\sigma\omega}}$ Skin depth

$\vec{E} \propto e^{-\frac{\hat{k}\cdot\vec{x}}{\delta}}$

Propagation in a conductor ($\omega \ll \gamma_0$)

Assume $\hat{k} \cdot \hat{x}$ is z.

$$\vec{b} = \vec{b}_0 e^{-\frac{z}{\delta}} e^{i\left(\frac{z}{\delta} - \omega t\right)}$$



Propagation in a conductor ($\omega \ll \gamma_0$)

$$\vec{H} = \frac{\hat{k} \times \vec{E}}{Z} \qquad Z = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu}{\epsilon_r + i\frac{\sigma}{\omega}}}$$

Good conductor. $\sigma >> \omega\epsilon_r$

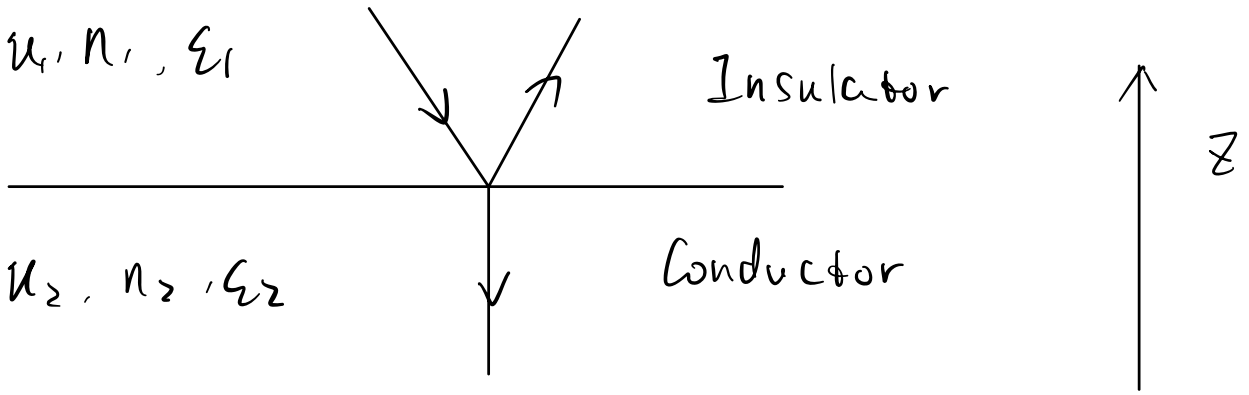
$$Z = \sqrt{\frac{\mu}{i\sigma/\omega}} = \sqrt{\frac{i\mu\omega}{\sigma}} \cdot \sqrt{-i}$$

$$= \sqrt{\frac{\mu\omega}{\sigma}} \cdot \sqrt{e^{-i\frac{\pi}{2}}} = \sqrt{\frac{\mu\omega}{\sigma}}\, e^{-i\frac{\pi}{4}}$$

$$\vec{H} = \hat{k} \times \vec{E} \sqrt{\frac{\sigma}{\mu\omega}}\, e^{i\frac{\pi}{4}}$$

45° phase diff between $\vec{H}$ and $\vec{E}$

Reflection and Transmission at the surface of a conductor ($\omega \ll \gamma_0$)



Snell's Eq

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

$$\sin\theta_2 = \frac{n_1}{n_2} \sin\theta_1$$

$$\simeq 0 \quad \Rightarrow \quad \theta_2 = 0$$

Fresnel Equations ($\omega \ll \gamma_0$)

Assume $\mu_1 = \mu_2$

$$r_p = \frac{n_2 \cos\theta_1 - n_1 \cos\theta_2}{n_2 \cos\theta_1 + n_1 \cos\theta_2}$$

θ_1 incident angle

θ_2 transmitted angle

Assume, $\theta_1 = 0$. Good conductor . $\frac{b}{w} >> \varepsilon_r$

$$r_p = \frac{n_2 - n_1}{n_2 + n_1} \qquad R_p = r_p^2 = \left| \frac{n_2 - n_1}{n_2 + n_1} \right|^2$$

Example: What is the reflection coefficient in a good conductor at normal incidence? ($\omega \ll \gamma_0$)

$$R_p = \left| \frac{n_r + i n_i - n_1}{n_r + i n_i + n_1} \right|^2 \overset{n_r = n_i}{=} \frac{(2n_r^2 - n_i^2)^2 + 4n_1^2 n_r^2}{[(n_r + n_1)^2 + n_r^2]^2}$$

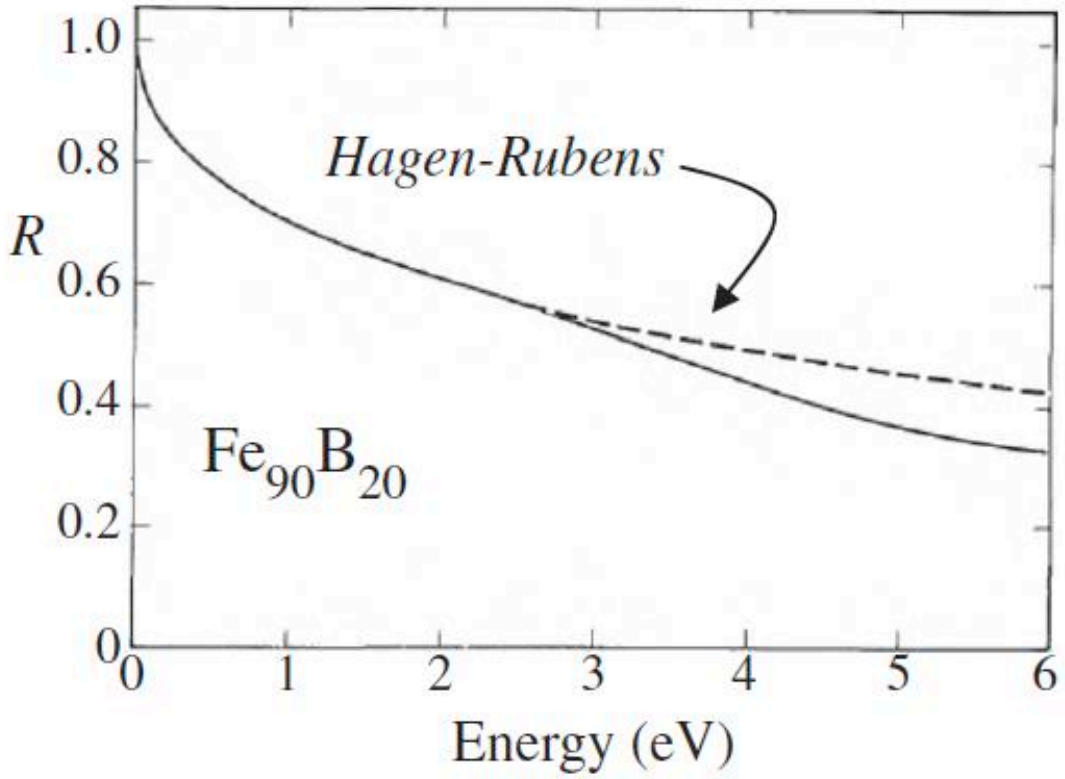
$$= \left(1 - \frac{2n_r n_1}{2n_r^2 + n_i^2 + 2n_r n_1}\right)^2$$

$$= 1 - \frac{4n_r n_1}{2n_r^2 + n_i^2 + 2n_r n_1} \simeq 1 - \frac{4n_r n_1}{2n_r^2} = 1 - \frac{2n_1}{n_r}$$

$$= 1 - \sqrt{\frac{\delta \varepsilon_1 \omega}{6_2}}$$

Hagen-Rubens Relation ($\omega \ll \gamma_0$)

Shiny metal



Brewster's angle ($\omega \ll \gamma_0$)

P polarization

$$r_p = \frac{n_2 \cos\theta_1 - n_1 \cos\theta_2}{n_2 \cos\theta_1 + n_1 \cos\theta_2}$$

$r_p = 0$. $n_2 \cos\theta_1 \neq n_1 \cos\theta_2$

Complex real

Brewster's angle ($\omega \ll \gamma_0$)

